

Qiskit Fall Fest 2025

Qiskit Beginner Challenge, plus Question Revisits

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Qiskit Advocate, Tier 1

Seattle Quantum Computing Meetup, Founder



Agenda

Announcements

Q News – Industry & Academia

Revisit Previous Questions

Qiskit Challenge Beginner Notebook

Talk about the next Meetup

Mingle





« [Darkness over America](#)

[The QMA Singularity](#) »

HSBC unleashes yet another “qombie”: a zombie claim of quantum advantage that isn’t

Today, I got email after email asking me to comment on a new [paper from HSBC](#)—yes, the bank—together with IBM. The paper claims to use a quantum computer to get a 34% advantage in predictions of financial trading data. (See also blog posts [here](#) and [here](#), or numerous popular articles that you can easily find and I won’t link.) What have we got? Let’s read the abstract:

Claims

“When and how will quantum computing broadly benefit humanity? Despite exhilarating recent progress, [we still don’t know](#). Here my friend Jens Eisert and I assess the current status and the challenges ahead. We are optimistic about the quantum future, but there’s a lot of work to do.”
John Preskill, Caltech.

<https://lnkd.in/gxuzxhj3>
2510.19928

What’s Needed

“Gaps that we still have to cross in order to reach broadly useful technology. An important take-away: **the biggest gap is not in hardware, but in developing the right quantum algorithms.**”
Koen Groenland, QuSoft.

Mind the gaps: The fraught road to quantum advantage

Jens Eisert^{1,2,3} and John Preskill^{4,5}

¹*Dahlem Center for Complex Quantum Systems, Freie Universität Berlin, 14195 Berlin, Germany*

²*Helmholtz-Zentrum Berlin für Materialien und Energie, 14109 Berlin, Germany*

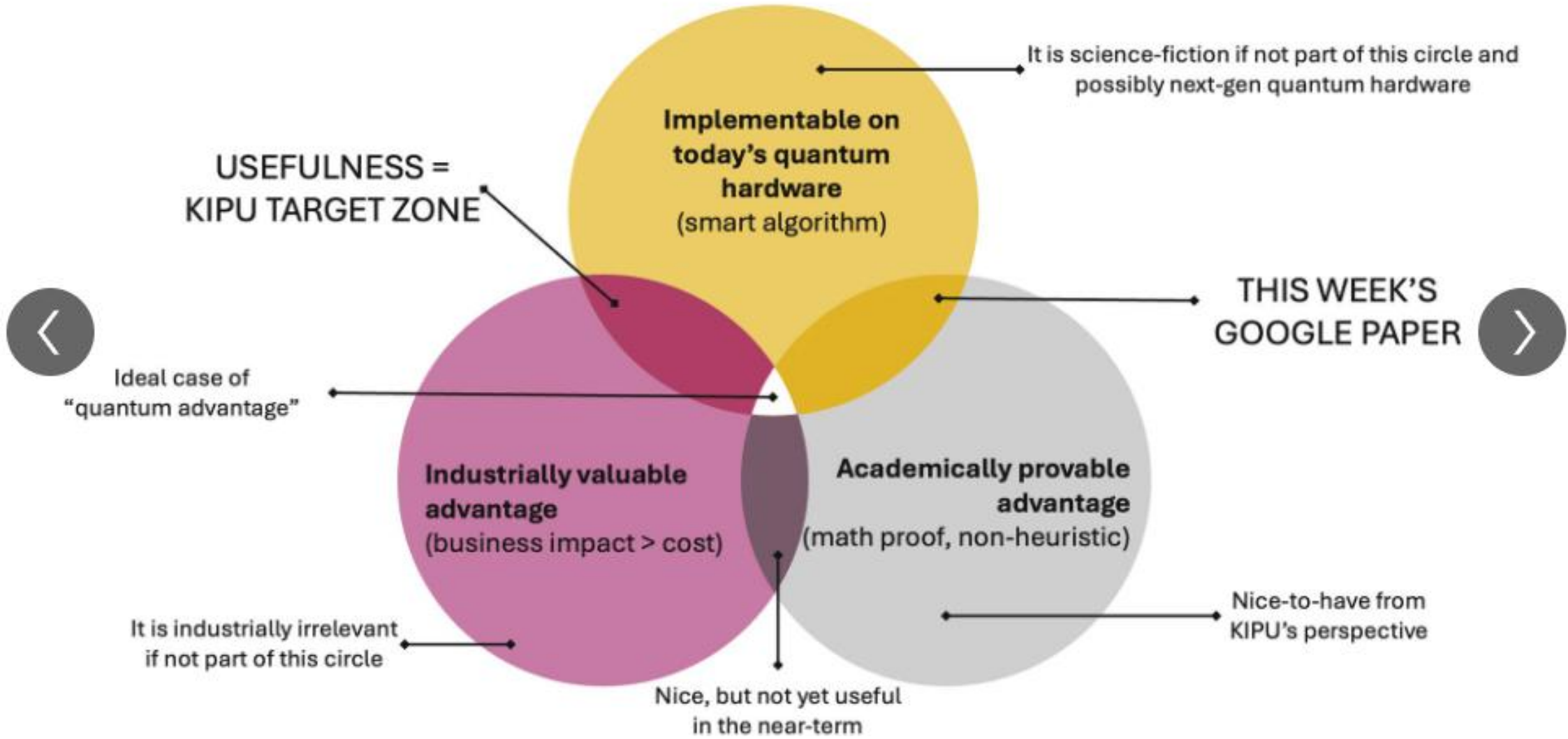
³*Fraunhofer Heinrich Hertz Institute, 10587 Berlin, Germany*

⁴*Institute for Quantum Information and Matter, California Institute of Technology, CA 91125, USA*

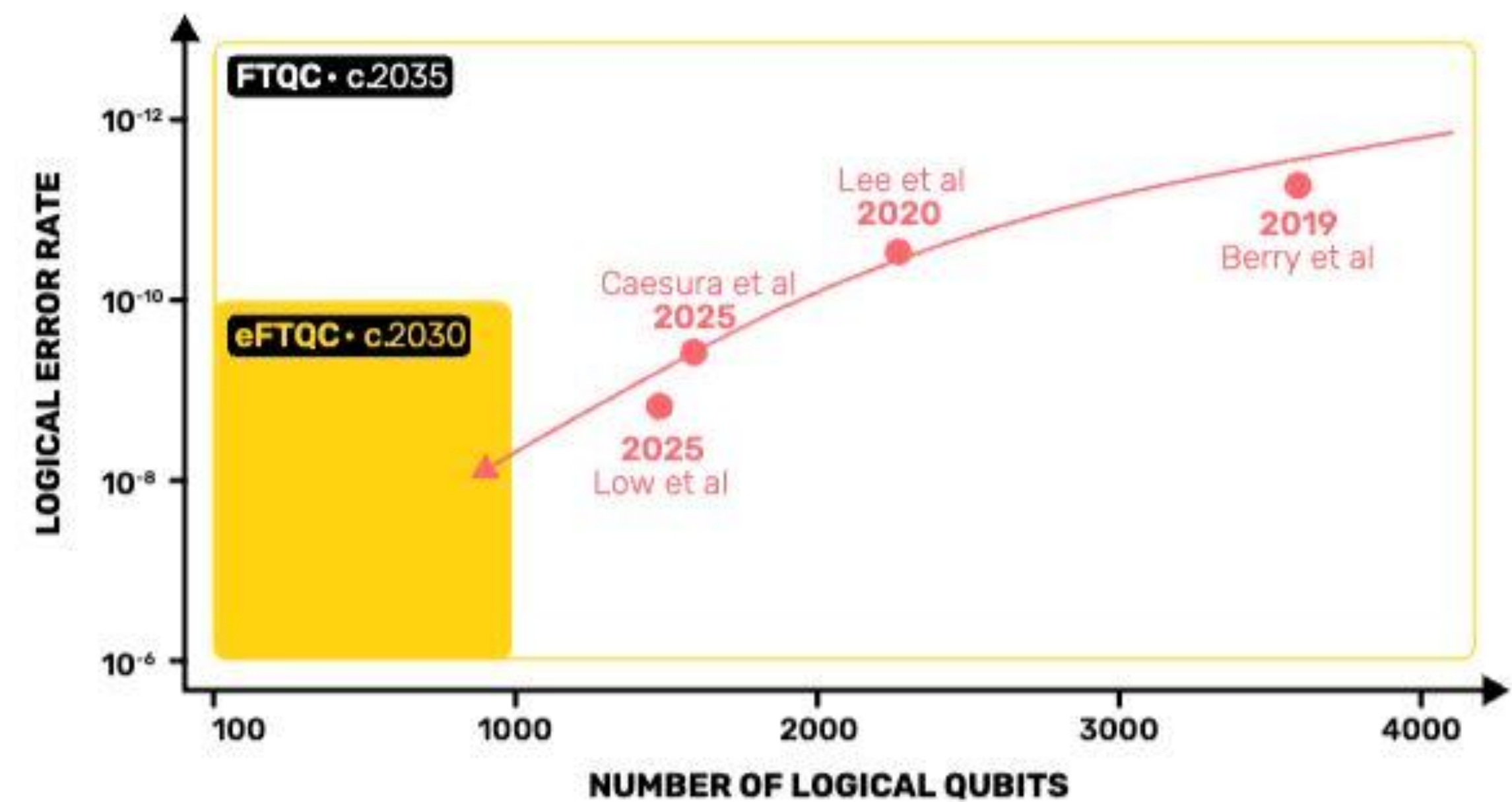
⁵*AWS Center for Quantum Computing, Pasadena, CA 91125, USA*



The Venn Diagram of “Quantum Advantage”



Quantum resources requirements for FeMoco



Feature	Automatski	IBM	Google	Microsoft	IonQ	Rigetti	Atom Computing	PsiQuantum	Quantinuum
Qubit Technology	Photonic	Superconducting	Superconducting	Topological	Trapped Ions	Superconducting	Neutral Atoms	Photonic	Trapped Ion
#Qubits (Physical)	Undisclosed. Only focus on Logical Qubits	127 (eagle) 433 (Osprey) 1,121 (Condor) 133 (Heron)	53 (Sycamore) 105 (Willow)	Claim 8 (Disputed)	20 (Aria) 32 (Forte)	84 (Ankaa-3)	1,180	???	56
#Qubits (Logical)	5000+ (Available For Trials)	??? Refuses to Publish Benchmarks	??? Maybe 2	N/A	???	???	24	???	23
Gate Depth	50,000	5,000	???	N/A	800	???	???	???	4900
Encoding	Discrete	Discrete	Discrete	Discrete	Discrete	Discrete	Discrete	Continuous Variable	Discrete
Temperature	Cryogenics (Photon Detectors)	Cryogenics	Cryogenics	Cryogenics	Room temperature + laser cooling to μ K	Cryogenics	Room temperature + laser cooling to μ K	Cryogenics (Photon Detectors)	Room temperature + laser cooling to μ K
Coherence	> 1 hour extremely long coherence since photons don't decohere thermally	300 μ s	100 μ s	N/A	> 1 hour	20 μ s	> 40 secs	10's of secs extremely long coherence since photons don't decohere thermally	Several Minutes
Fidelity	99.999%	99.9%	99.9%	N/A	99.9%	99.9%	99.6%	99%	99.99%
Year Of Incorporation	1993	1911	1998	1975	2015	2013	2018	2016	2021
Year Of Launch	1997	2016	2019	Not Yet Launched. Just an Unproven Concept.	2018	2017	2021	Not Yet Launched	2021



Announcements



1

Submissions Deadline for earning a Winning Certificate is **Extended** to **Monday, Nov 24!!!**

The **non-coding and Hackathon options** can be starts on **projects**; they don't have to be fully worked out.

2

Project Idea: (1-2 pgs max total)

Short answers on questions:

- How will QC affect our futures?
- How will QC and AI affect our futures?
- What are the pro's and con's of each type of qubit?
- How many qubits are required for Shor's alg?

3

Our **last in-person FF25 Meetup** will be next **Saturday, Nov 1, 2-4 pm**, at the Montlake Branch of the SPL.

Topics TBD:

- 1) DiVincenzo Intermediate Notebook
- 2) Working in QC, Getting a job, Where are the jobs?

4

Upcoming Summit:

TechAlliance Policy Matters Summit | The Quantum Frontier

<https://luma.com/sxwd8icd>

TechAlliance Policy Matters Summit | The Quantum Frontier

Overview -

Washington has what it takes to lead in quantum top-tier research, major tech players, and a growing innovation ecosystem. What's missing is a coordinated strategy. The 2025 Policy Matters Summit will bring together policymakers, educators, industry, and startups to launch a shared vision for Washington's quantum future.

Summit Goals -

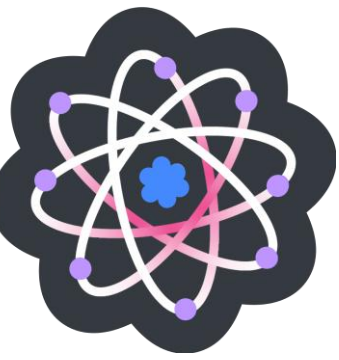
Position Washington as a hub for quantum R&D, talent, and commercialization
Spark collaboration and follow-on action statewide

Why Now -

States like Texas, Illinois, and Maryland are moving fast - passing laws, building hubs, and attracting funding. Washington has the assets. This Summit is our chance to catch up - and take the lead.

Summit Agenda

Wed, Nov 19, 2025



8:30 Breakfast/Networking

9:00 Opening remarks by **Lieutenant Governor Denny Heck**

9:20 Quantum Basics - A beginner-friendly overview of quantum concepts, technologies, and real-world applications

11:00 Q-Day 101 - What is Q-Day and why does it matter? Learn how quantum will impact cybersecurity, infrastructure, and the economy.

12:00 Networking lunch

1:00 The Quantum Landscape – National and Global Initiatives - Explore how other states and countries around the world are advancing quantum

2:00 Building a Quantum Ecosystem in WA

Others Are Moving Fast

- passing laws
- building hubs
- attracting funding

Look at IEEE Quantum Week Locations:

- 2022 Broomfield, CO
- 2023 Seattle
- 2024 Montreal
- 2025 Albuquerque
- 2026 Toronto

<https://qce.quantum.ieee.org/2025/>

IL

Chicago Quantum Summit, Nov 3-4

<https://chicagoquantum.org/summit2025>

MD

In 2025, **Maryland's quantum ecosystem** is experiencing significant growth, fueled by major **investments, partnerships** with technology leaders like Microsoft and DARPA, and the continued expansion of homegrown companies such as IonQ. (Google)

TX

Texas is solidifying its position as a major hub for quantum technology through the recently passed **Texas Quantum Initiative**, which aims to boost state investment and coordinate research and development. The effort involves collaborations between academic institutions, industry leaders, and state government. (Google) (**Abilene, Texas:** The first and flagship site is a massive AI data center \$500B Stargate Project.)

What is Q-Day?

The quantum computing meaning of Q-Day

- What it is:**

Q-Day is the day a powerful enough quantum computer can use algorithms like [Shor's algorithm](#) to crack asymmetric encryption that protects data like online banking, email, and other secure communications.

- Why it's a concern:**

It would create a "cryptographic apocalypse," where sensitive data could be easily decrypted by attackers, impacting everything from financial transactions to secure messaging.

- Current status:**

Experts debate the exact timeline, but they agree that organizations need to prepare by transitioning to [post-quantum cryptography](#) (PQC) to stay secure.

- Preparing for Q-Day:**

The race is on to develop and implement new quantum-resistant encryption methods before a powerful quantum computer arrives.

The historical meaning of Q-Day

- What it is:** In 1945, "Q-Day" was the name of the dress rehearsal for the Trinity atomic bomb test, the first detonation of a nuclear weapon.

5 Misconceptions about Q-Day

By Werner Coomans

24 May 2025

<https://www.nokia.com/quantum/5-misconceptions-about-q-day/#:~:text=There%20are%20two%20families%20of,which%20are%20currently%20undergoing%20standardization.>



Revisit Previous Questions



1

Greek Alphabet

2

Data and Quantum Computers

- Data Loading Project
- Watch out for Bad Articles, written by AI or humans
- Small Data, Big Solution Space (Medium: J Krupansky)
- Now-ish vs. Later-ish (FTQC, qRAM)

3

AI + Quantum, Quantum + AI

- Qiskit Code Assistant
- AI Software Survey (TensorFlow?)
- Quantum Software Survey (TensorFlow Quantum?)

4

Computer Chips

- How SC qubits work, Qiskit Channel, Zlatko's talk, Parts 1 and 2, Lectures 16-21, (~6 hrs)
- How QPUs and CPUs are made

Greek Alphabet

Phi and Psi are used a lot, also Alpha, Beta, Pi, and Theta, and, well, the others too!

https://en.wikipedia.org/wiki/Greek_alphabet

Αα	Ββ	Γγ	Δδ	Εε
Alpha	Beta	Gamma	Delta	Epsilon
al-fah	bay-tah	gam-mah	del-tah	ep-si-lon
Ζζ	Ηη	Θθ	Ιι	Κκ
Zeta	Eta	Theta	Iota	Kappa
zay-tah	ay-tah	thay-tah	eye-o-tah	cap-ah
Λλ	Μμ	Νν	Ξξ	Οο
Lambda	Mu	Nu	Xi	Omicron
lamb-dah	mew	new	zz-eye	om-e-cron
Ππ	Ρρ	Σσς	Ττ	Υυ
Pi	Rho	Sigma	Tau	Upsilon
pie	roe	sig-mah	taw	oop-si-lon
Φφ	Χχ	Ψψ	Ωω	
Phi	Chi	Psi	Omega	
fie	k-eye	sigh	o-may-gah	



AI and Quantum Computers



Qiskit Code Assistant

- IBM Quantum has one
- Available via a Premium Plan
- Designed to create Qiskit 2 code
- <https://quantum.cloud.ibm.com/docs/en/guides/qiskit-code-assistant>

AI Software Survey

- O'Reilly Tech Trends Report (from Jan 6 '25: based on '24, '23)
- Look for TensorFlow
- <https://www.oreilly.com/pub/pr/3465>

Quantum Open-Source Software Survey

- See the Unitary Foundation's annual survey
- <https://unitaryfoundation.github.io/survey-2024/>
- Look for TensorFlow Quantum in the results
 - Look for CUDA-Q

AI + Q, Q + AI

- AI helping Quantum
- Quantum helping AI
- Conference Q+AI
- Company: e.g. Sandbox AQ
- IEEE Quantum Week 2025, *can still register for on-demand content (~\$200)*
- <https://web.cvent.com/event/3f8c3754-761c-4db5-a3f5-20003fcc785c/summary>

AI Tool Trends

Technology Trends for 2025
Report by O'Reilly, Jan 6 2025

How to Choose Between
TensorFlow vs PyTorch in 2025

<https://www.niso.org/niso-io/2025/01/oreilly-offers-insights-tech-trends-2025#:~:text=Usage%20of%20material%20about%20Software%20architecture.>

“PyTorch is easier to use and better for research, while TensorFlow is better for production and scalability.”

<http://youtube.com/watch?v=yOGi4vmtNaY>

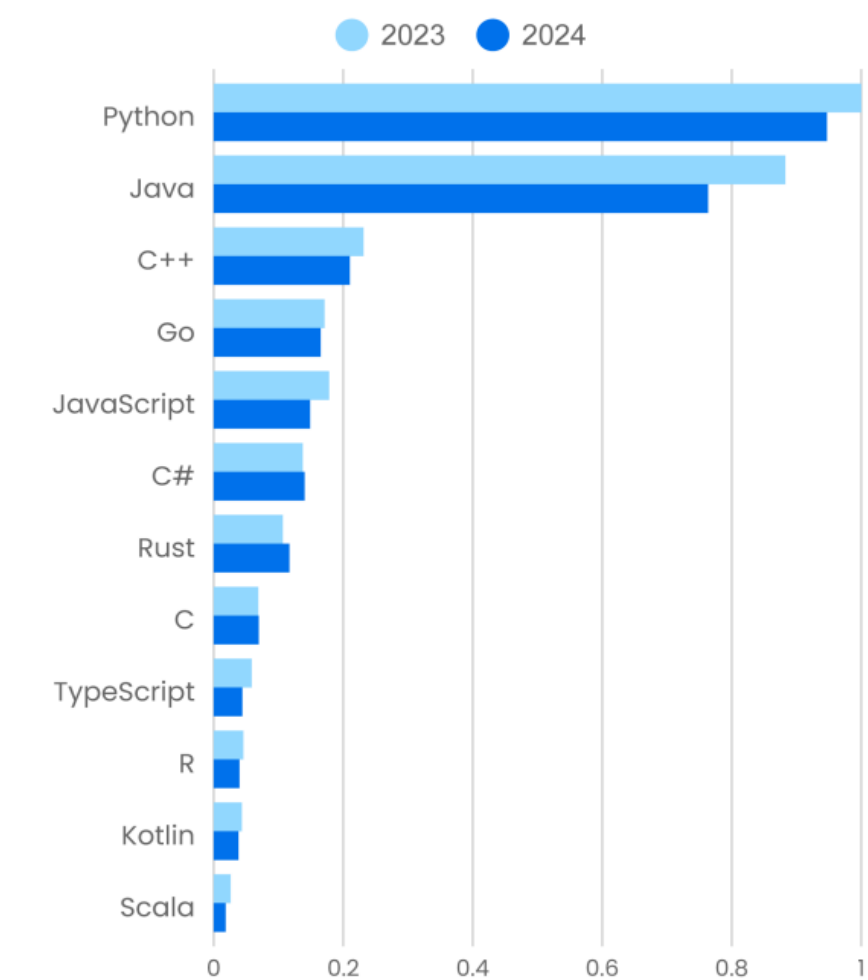
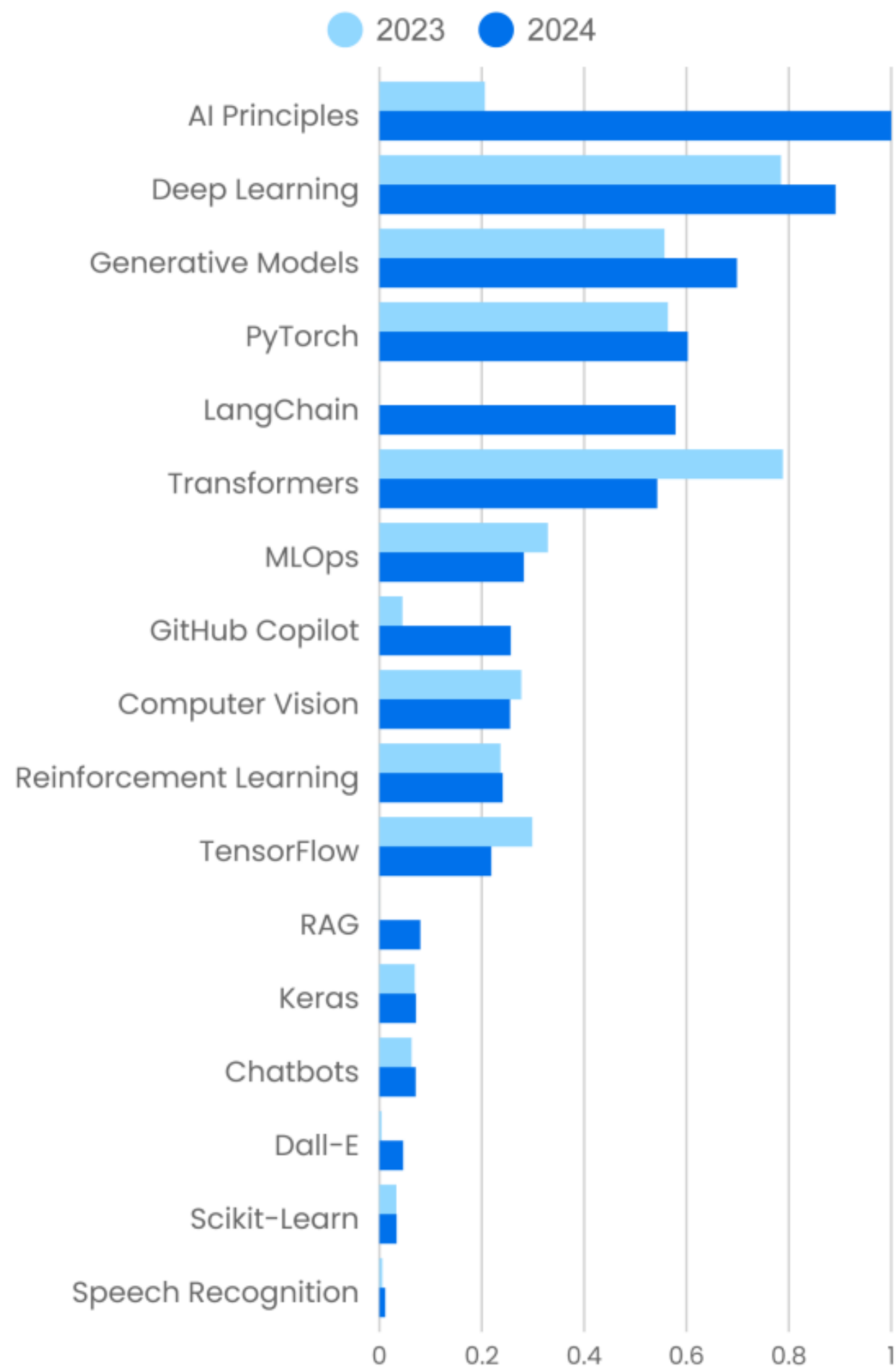


Figure 5. Programming languages

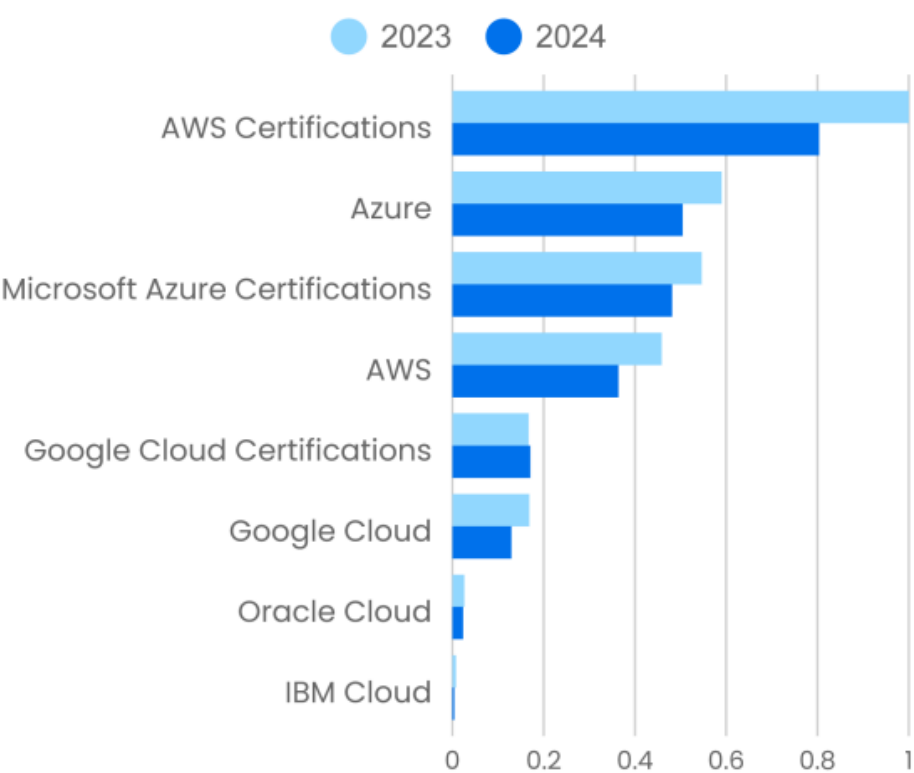


Figure 9. Cloud providers and certifications

Figure 2. Skills needed for AI



Panorama of the Open-Source Large Model Ecosystem 2.0

In Ant Group’s newly released *Panorama of the Open-Source Large Model Ecosystem 2.0*, TensorFlow has been officially removed.

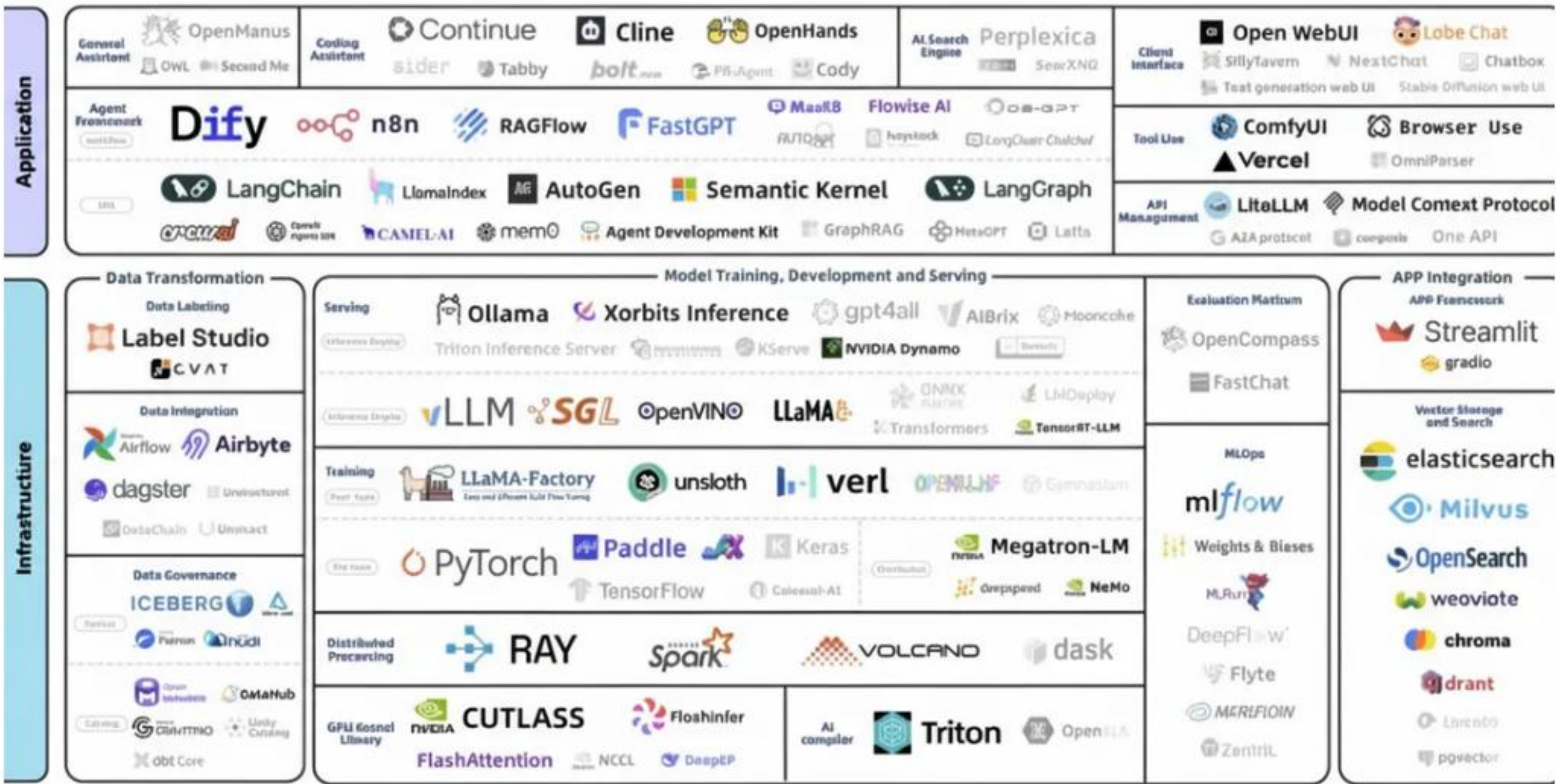
Medium article, *TensorFlow Is Dead. PyTorch Won.*, [Pan Xinghan](#), Sep 15, 2025

How Industry: Cracks in TensorFlow’s Fortress

“In industry, TensorFlow still hangs on, thanks to its early enterprise adoption. Big companies with legacy models — think Google Translate, some cloud pipelines, some banks — still run it. But the cool kids aren’t choosing it anymore.

OpenAI trained GPT-3 and DALL·E on PyTorch. Tesla’s Autopilot runs on PyTorch. Meta, Microsoft, and Amazon have rallied behind it. NVIDIA and Intel are optimizing PyTorch first. Even Google Cloud now openly supports PyTorch and quietly shifts research toward JAX instead of TensorFlow.

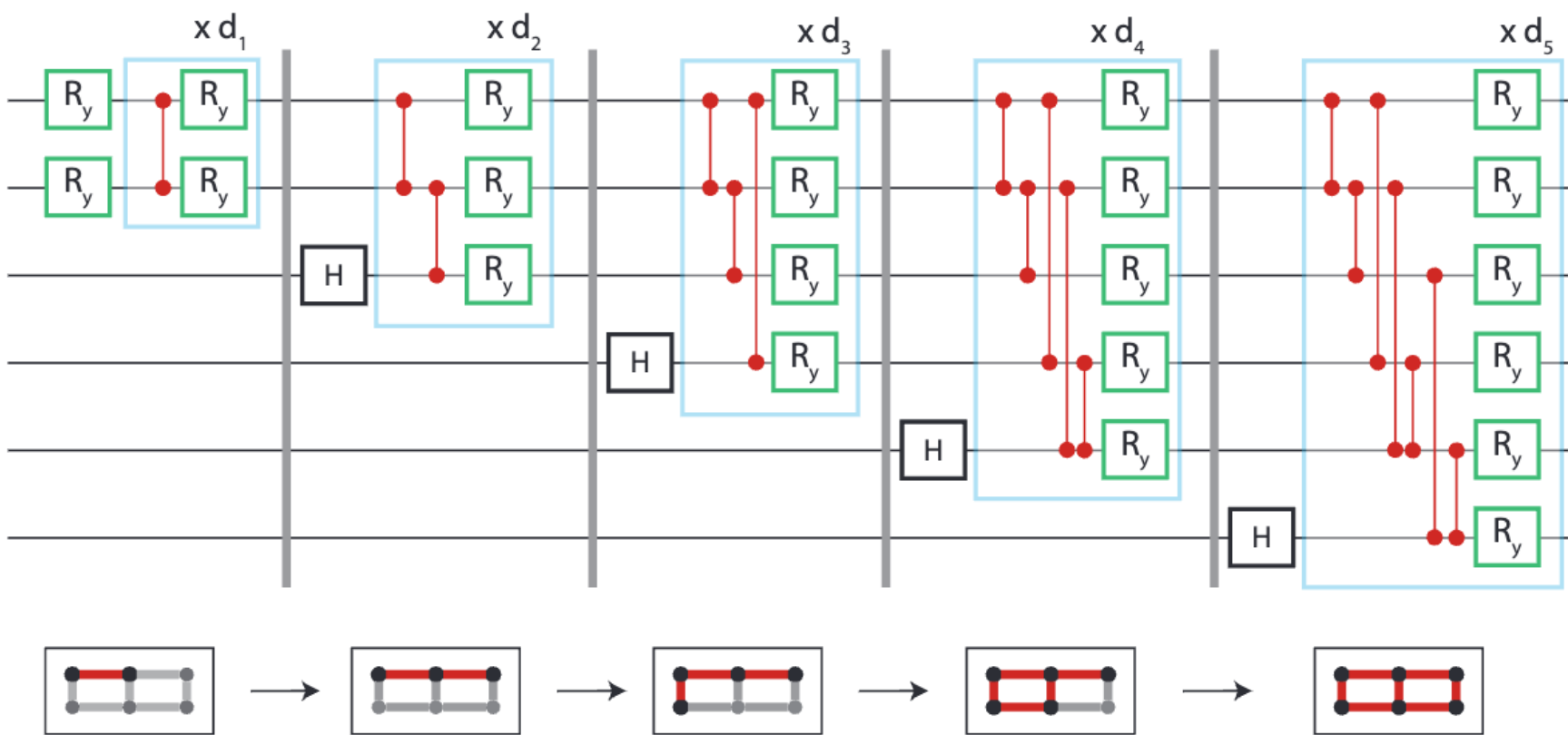
Yes, TensorFlow has stars on GitHub. But stars don’t equal momentum. Forks, contributors, and new repos overwhelmingly lean PyTorch. The trend lines are clear: TensorFlow is legacy, PyTorch is future.”



Panorama of the Open-Source Large Model Ecosystem 2.0 released by Ant Group



Data and Quantum Computers



Blue Qubit, 2311.12929 <https://www.bluequbit.io/quantum-data-loading>



Figure 1: A circuit ansatz with hierarchical architecture on 6 qubits laid out on a grid. Vertical gray lines on the circuit indicate a partial circuit which is variationally learned before using it as input to the next sub-circuit.

Quantum State Preparation

Phys. Rev. Lett. **129**, 230504 – **Published 30 November, 2022**

“Quantum state preparation is an important subroutine for quantum computing. We show that any n -qubit quantum state can be prepared with a $\Theta(n)$ -depth circuit using only single- and two-qubit gates, although with a cost of an exponential amount of ancillary qubits.”

Little Data with Big Solution Space

“No, quantum computers are not appropriate for *big data* problems. Rather, they are best for problems with a fairly small amount of data which has a very large solution space — so called *combinatorial explosions*. So, rather than call it *Big Data*, I call it *Little Data* with a *Big Solution Space*. This informal paper introduces the notions of *Little Data with a Big Solution Space*.”

<https://jackkrupansky.medium.com/little-data-with-a-big-solution-space-the-sweet-spot-for-quantum-computing-b6c6d76ada6d>

Quantum Resource Estimation for Data Loading or State Preparation in Quantum Computers

Google AI Overview

“Resource estimation for data loading and state preparation is a critical aspect of analyzing the overall cost of a quantum algorithm. The resources required often scale exponentially with the number of qubits for preparing a generic quantum state. However, tailored approaches can offer substantial reductions in circuit depth and qubit requirements for specific problems.

Key resource metrics include the number of qubits, quantum gates, and execution time. This process is particularly challenging because it involves translating classical data into a quantum format.

Encoded classical data: In quantum machine learning (QML) and other data-driven algorithms, classical data must be loaded into a quantum state. The efficiency of this process is a major bottleneck.”

Data and Quantum Computers



Qiskit Functions

<https://quantum.cloud.ibm.com/functions>

Multiverse Computing

Singularity Machine Learning Function

“With the Singularity Machine Learning function, you can solve real-world classification problems on quantum hardware without requiring quantum expertise. This function, based on ensemble methods, is a hybrid classifier which combines classical methods like boosting for ensemble training with quantum methods like quantum approximate optimization (QAOA) for optimizing the trained ensemble for maximum diversity, and generalization. Unlike other quantum machine learning solutions, this function is capable of handling large-scale datasets with millions of examples and features without being limited by the number of qubits. It is also highly flexible, and can be used to solve classification problems across a wide range of domains, including finance, healthcare, and cybersecurity.”

Quantum Resource Estimation

[Quantum Computing Report, article, Update on the GQI Quantum Resource Estimator Playbook](#)

<https://quantumcomputingreport.com/update-on-the-gqi-quantum-resource-estimator-playbook/>

[Resource Estimation Tools](#), Google AI Overview

Tools for quantum resource estimation include **Azure Quantum Resource Estimator**, **QREChem**, **TopQAD**, and **PsiQuantum's QREF and Bartiq**. These tools help estimate the logical and physical qubits, runtime, and other resources needed for a quantum program on a fault-tolerant quantum computer by taking into account various parameters like hardware specifications and error correction codes.

[What is QRE and why is it hard?](#)

<https://www.mustythoughts.com/perfect-qre-sw>

Superconducting Qubits

Superconducting Qubits I: Quantizing a Harmonic Oscillator, Josephson Junctions - Part 1, Lecture 16, by Zlatko Minev, IBM Quantum (now at Google)

<https://www.youtube.com/watch?v=kV0RsKXSqRg>

Also, see Lectures 17-21, approx. 6 hours in total



Zlatko has several presentations on noise too. One example,

<https://www.zlatko-minev.com/blog/quantum-error-mitigation-bss23>

How are QPUs Fabricated?

Inside Quantum Chip
Manufacturing: From Qubits to Fabrication, SpinQ Blog

<https://www.spinquanta.com/news-detail/inside-quantum-chip-manufacturing-from-qubits-to-fabrication>

How To Build A Quantum Computer

YT Channel: Lukas’s Lab (9 min)
<https://www.youtube.com/watch?v=N06hC1GL1ns&t=421s>

Intel’s Big Bet: Inside the Chipmaker’s Make-or-Break Factory

<https://www.nytimes.com/2025/10/09/technology/intel-chip-factory-fab52.html?searchResultPosition=1>

How Semiconductors Are Fabricated

<https://www.nytimes.com/2022/04/08/technology/intel-chip-shortage.html>

QC Biggest Challenge?



Zlatko says, “Noise.” (2022)

<https://www.zlatko-minev.com/blog/pec-talk>

How Deal with Noise?

Depends on the type of noise.

Is it possible to learn the noise with accuracy, efficiency, and scalability?

$Q_1 \dots$
 $Q_2 \dots$
 $\vdots$
 $Q_n \dots$

$\tilde{\Lambda}_i$ U_i

Energy relaxation T_1
Dephasing T_2
Coherent errors ZZ
Classical crosstalk
Quantum crosstalk
State preparation error
Measurement correlated errors
...

Control errors
Photon shot noise
1/f charge noise
1/f flux noise
Nonequilibrium quasiparticles
Leakage
Cosmic rays
...

Zlatko Minev, IBM Quantum (28)

How to deal with errors due to noise?

Monitor
Error occurs
Error detect

Quantum error correction

Shor, PRA (1995), ...

Monitor
Error anticipated
Tell signal detected

Catch and reverse

Minev, Nature (2019), ...

No monitor
Error occurs
Error undetected

Error mitigation

... subject of today

Zlatko Minev, IBM Quantum (5)



Qiskit Beginner Challenge Notebook



1

Software Setup

2

Circuits and Visualization

3

Optimize, then Run on
Simulator or Fake Backend

4

Run on Real Quantum HW

Qiskit Fall Fest 2025

Thanks for Coming!!!

Deadline for Submissions: Monday Nov 24, 10 am PT

New Zoom Meetup: Tuesday Oct 28, 6 pm PT

Next IRL Meetup: Saturday Nov 1, 2 pm PT

Upcoming Event: TechAlliance, Nov 19, South Seattle

Upcoming Meetup: Date? in December w/**Holiday Party**



Century of Quantum

